

LiDAR to Mesh using Neural Kernel Surface Reconstruction

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1. Introduction

Indoor mobile LiDAR enables fast capture of building-scale environments, but downstream applications (e.g., digital twins, inspection, simulation, XR/visualization, and robotics) typically require an explicit *surface* representation rather than raw points. Turning LiDAR scans into high-quality meshes is difficult because real scans exhibit noise, strongly varying point density, and missing regions due to occlusions; at the same time, datasets can contain millions of unstructured points, making scalability (memory/runtime) a first-class concern. These challenges are well known in surface reconstruction from real-world point clouds.

Recent learning-based implicit reconstruction methods can improve robustness and detail, but many approaches struggle to scale cleanly to very large scenes. Neural Kernel Surface Reconstruction (NKSr) addresses this by combining learned kernels with compact support and sparse linear solving, targeting fast, large-scale reconstruction from sparse and noisy point clouds [2].

2. Problem Statement and Research Questions

The goal of this project is to build and evaluate a practical pipeline that converts large-scale indoor LiDAR point clouds captured with the NavVis VLX¹ scanner into high-quality mesh. In contrast to small academic benchmarks, NavVis captures building-sized scenes with millions of points, uneven sampling density, and occlusions, which makes both reconstruction quality and computational scalability central requirements. To address these challenges, we adopt Neural Kernel Surface Reconstruction (NKSr) as the core meshing method. NKSr is designed to reconstruct surfaces from sparse and noisy point samples while remaining efficient through compactly supported kernels and sparse computation. Rather than re-implementing NKSr, we focus on turning it into a usable large-scale pipeline: we integrate NKSr with fVDB [6] sparse voxel data structures to enable chunked processing, memory-efficient spatial indexing, and controllable resolution/runtime trade-offs on real

NavVis VLX scans.

We therefore define the following research questions for this project work:

1. **Scalability:** Can NKSr reconstruct meshes for large, complex indoor LiDAR scans within practical runtime and memory limits, and how does performance scale with scene size?
2. **Quality/robustness:** Compared to TSDF-based baselines, does NKSr produce higher-quality meshes on real mobile mapping data (e.g., fewer artifacts/holes and better detail) under noise, varying density, and occlusions?

3. Related Work

Our pipeline is built around *Neural Kernel Surface Reconstruction* (NKSr) [2], a method for reconstructing surfaces from oriented point clouds by combining learned priors with a scalable variational formulation. NKSr is closely related to *Neural Kernel Fields* (NKF) [5], which introduced the idea of learning a kernel representation and recovering the implicit surface via a regression-style solve. While NKF demonstrates strong reconstruction quality, its dense kernel formulation limits scalability. NKSr addresses this limitation by defining the kernel basis on a hierarchical sparse voxel structure, which makes the resulting system sparse and enables reconstruction on scenes with millions of points while improving robustness to sensor noise.

To handle building-scale indoor scans efficiently, we rely on fVDB [6], a GPU-oriented framework for learning and computation on hierarchical sparse voxel grids. fVDB provides efficient grid construction, sparse convolutional operations, and traversal primitives on its IndexGrid representation. In our setting, it serves as the practical backbone for processing NavVis VLX point clouds and running NKSr at large scale under realistic memory and runtime constraints.

Neural implicit reconstruction has also been explored in LiDAR-centric settings that go beyond pure point-cloud surface fitting. For example, *Neural Surface Reconstruction and Rendering for LiDAR-Visual Systems* [3] combines sparse LiDAR with posed imagery to improve completeness via learned occupancy/SDF representations, and *Neural Implicit Representation for Highly Dynamic LiDAR*

¹<https://www.navvis.com/vlx-2>

Mapping [7] studies reconstruction in dynamic environments. These works highlight the growing interest in neural implicit for practical mapping, and motivate evaluating whether NKSR can provide similar benefits for large indoor scans when used as a drop-in reconstruction module.

Finally, we position our evaluation against established surface reconstruction baselines and prior learning-based approaches. The survey and benchmark by Sulzer *et al.* [4] provides a systematic overview of classical methods such as Screened Poisson Surface Reconstruction and TSDF fusion, together with common evaluation metrics for geometric accuracy and robustness. Among learning-based baselines, POCO [1] is a strong recent method for point-cloud surface reconstruction. NKSR reports improved accuracy and generalization over such competitors across multiple datasets, particularly under noise and sparsity, making it a suitable foundation for our LiDAR-to-mesh pipeline.

4. Methodology and Approach

4.1. Data Acquisition

We will acquire large-scale indoor point clouds using a NavVis VLX2 mobile mapping system. In addition to the LiDAR measurements, the device’s imagery will be recorded to enable RGB-colorization of the point clouds and to support visual inspection of reconstruction results. Georeferencing will be established via the existing control point network, ensuring a consistent coordinate frame across all acquisitions. To minimize drift in the inertial SLAM trajectory estimation, the acquisition plan will include loop closures (i.e., closed trajectories) and repeated traversals of key areas.

All datasets will be captured in the ETH Zurich HG/CAB/CNB buildings.

If time permits, we will additionally collect reference datasets with survey-grade terrestrial laser scanners such as the Leica BLK360² and Leica RTC360³ to directly compare how sensor characteristics and acquisition workflows impact reconstruction quality.

4.2. Data processing

NavVis devices include an integrated software package for point-cloud processing. Dataset cleaning will be performed using this software, while the core algorithmic logic will be implemented in Python. Given the computational complexity, the use of HPC clusters will be considered.

²<https://leica-geosystems.com/products/laser-scanners/scanners/leica-blk360>

³<https://leica-geosystems.com/products/laser-scanners/scanners/leica-rtc360>

4.3. Timeline and Milestones

A rough estimation of our planned timeline is provided in figure 1. Our plan is to begin with exploring existing approaches for mesh generation on available point clouds, before starting with our own scanning missions. Once the initial setup and exploration is completed and our first scans are acquired, we plan to simultaneously work on both processing pipeline and additional scans.

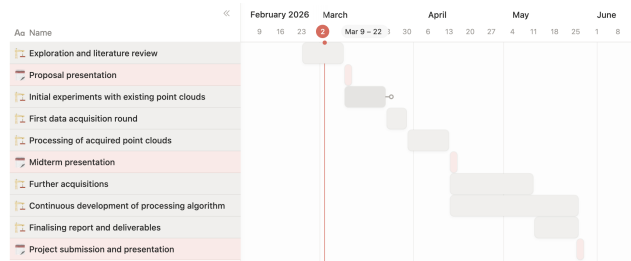


Figure 1. Estimated timeline

5. Expected Results

The following deliverables will be provided:

A functional codebase that implements an algorithm to efficiently compute a mesh based on LIDAR point cloud.

Comparative visualizations against other surface reconstruction algorithms (e.g., TSDF-based reconstruction using Marching Cubes, Poisson Surface Reconstruction, and Ball Pivoting Algorithm). Given the lack of ground truth data to be compared to, performance will be evaluated by comparing to other method’s resulting meshes.

References

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